

AI & Self-Service

To me, an AI & Self-service data platform is an analytics layer that abstracts the complexities of code and infrastructure from business stakeholders while still allowing them to gain quick and accurate insights and visualizations. The use of natural language querying is especially handy here for the comprehension of user requests, generation of code, designing and deployment of dashboards and reports.

There are a couple of factors that make it difficult to access insights within an organization:

1. Presence of data that naturally quickly decays e.g. customer complaints
2. Disparate data sources e.g social media, databases, payment platforms, logs
3. Lack of data governance and enforcement of data quality e.g using GreatExpectations in a Medallion Architecture
4. Failure to completely audit a company's data.
5. Lack of enough members in a data team to promptly synthesize insights from data.

A platform that automates text-to-sql, text-to-code processes using a language model can solve these problems by collecting data, synthesizing results, and generating reports.

Such a platform however creates and faces certain challenges:

1. Hallucination tendencies
Without a complete understanding of the system, it may generate incomplete insights or incorrect sql filters, joins, or groupings.
The explanations may also be fabricated since language models are naturally stochastic in nature.
2. Semantic Inconsistencies
Business definitions matter. For example, Sunculture has a PAYG which translates to Pay-As-You-Grow and this may not be immediately apparent to a language model unless explicitly stated.
3. Trust & Adoption
Owing again to the stochastic nature of LLMs, the result may not always be consistent which in some cases erodes trust from the users.

Experience & Examples

a) Experience with AI-Powered Analytics and Self-Service Data Platforms

Over the last couple of years I have actively worked on designing and building AI-driven analytics systems. These systems have been focused on allowing the use of natural language for automating the generation of insights, simplifying access to business intelligence and enabling automated generation of reports.

My most primary project has been Core&Outline, an AI-powered analytics and machine learning decision support system for organizations. In there I created two separate platforms: Atlas - The infrastructure background; Corine - The C-suite dashboard. Atlas is the centralized warehouse for data ingestion, analytics, predictive modeling, and AI-assisted decision-making across multiple business domains such as finances, customer analytics, SaaS metrics, and operational monitoring.

b. Role and Contribution

My role has primarily involved:

- Designing the analytics and AI architecture
- Building data ingestion and transformation pipelines
- Developing machine learning and AI systems
- Creating analytical warehouse schemas and feature stores
- Designing natural-language analytics workflows
- Developing proof-of-concept AI applications for business intelligence and operational automation
- Building interfaces for data exploration and insight generation

Although many components are still under active development and iteration, the platform has been designed with the long-term goal of democratizing access to advanced analytics and AI capabilities for both technical and non-technical users.

The project repos are all private which would make it difficult for me to share but here are some links to view the demos and understand the products I have built:

<https://youtu.be/G68d-VcjMvI>

https://www.linkedin.com/posts/tomitsuma_what-sets-coreoutline-as-the-decision-support-ugcPost-7458072229596004354-XngQ/?utm_source=share&utm_medium=member_desktop&rcm=ACoAACb5XNEBAXMxtEV3IXRQDzA4LL1qO0h3scs

<https://www.coreoutline.com/>

c. Proof-of-Concept (POC) Proposal

If I were to build a POC using the provided SunCulture dataset, I would focus on creating an AI-powered operational analytics assistant that enables business teams to interact with company data using natural language. I would design an analytical warehouse with fact tables defining, sales, installations, leads_funnels etc and data marts as well. I would then define ways for LLMs to safely interact with the data by creating a sunculture_ai user in the database with read-only RBAC defined. I have added a diagram to show how the interaction would be.

